Proxmox VE Migration

Day 1: Platform Comparison, Assessment & Preparation

Day 1 — Agenda

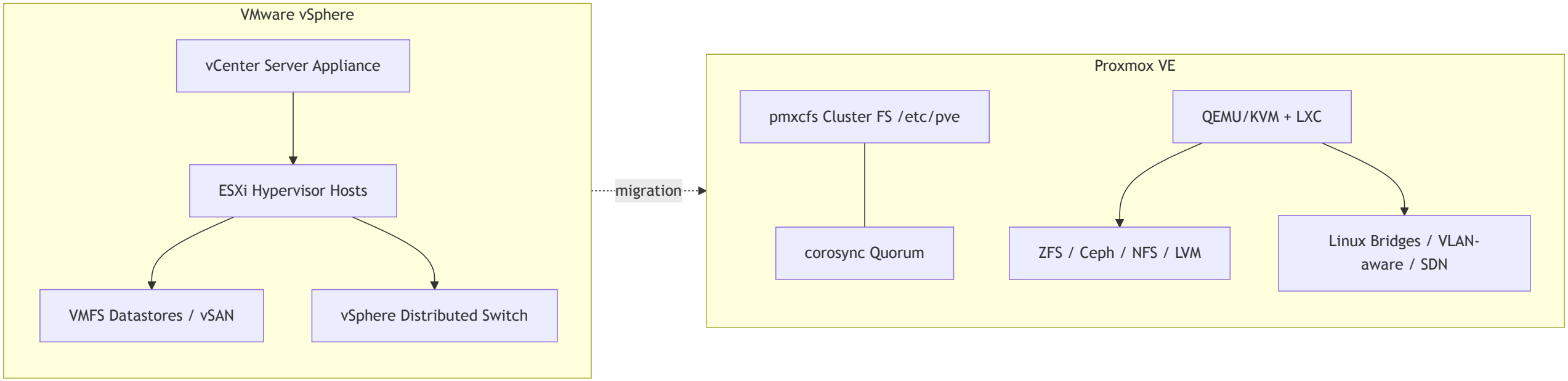
1. Course Introduction & Objectives
2. VMware vSphere vs. Proxmox VE — Platform Comparison
3. Hypervisors: ESXi, KVM & LXC
4. Management & Operating Models
5. Licensing, Support & TCO
6. Realistic Performance Expectations
7. Proxmox VE Web UI, CLI & SSH Access
8. Host, VM & Container Management — Limits of Containers
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1. Course Introduction & Objectives

This 3-day course teaches a **structured migration** from VMware vSphere to Proxmox VE: platform comparison, assessment, preparation, VM import, guest adaptation, testing, hardening, and production cutover.

By the end of the course you will be able to:

- Compare vSphere and Proxmox VE architectures, management models, licensing, and performance expectations
- Inventory a VMware environment and classify workloads for migration waves
- Design a Proxmox VE target for network, storage, identity, and backup
- Use the Proxmox VE import path for ESXi VMs (Linux, Windows, FreeBSD)
- Adapt guest operating systems (drivers, boot, network, storage) after import
- Validate migrated VMs with functional and performance tests


VMware vSphere
vCenter Server Appliance
ESXi Hypervisor Hosts
VMFS Datastores / vSAN
vSphere Distributed Switch
migration
Proxmox VE
pmxcfs Cluster FS /etc/pve
corosync Quorum
QEMU/KVM + LXC
ZFS / Ceph / NFS / LVM
Linux Bridges / VLAN-aware / SDN

2. VMware vSphere vs. Proxmox VE

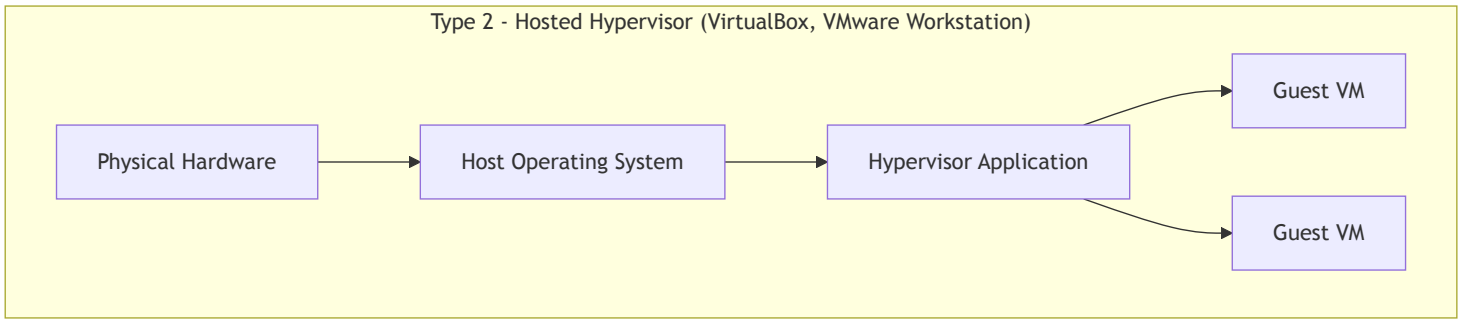
Area	VMware vSphere	Proxmox VE
Hypervisor	ESXi (VMkernel)	KVM via Linux kernel + QEMU
Management plane	vCenter Server Appliance	Distributed: web UI/API on every node, config in `/etc/pve`
Cluster state	vCenter database	`pmxcfs` (SQLite replicated by corosync)
VM storage	VMFS, vSAN, NFS, iSCSI	ZFS, LVM-thin, Ceph RBD, NFS/CIFS, directory
Networking	vSwitch / Distributed vSwitch	Linux bridges, VLAN-aware bridges, SDN

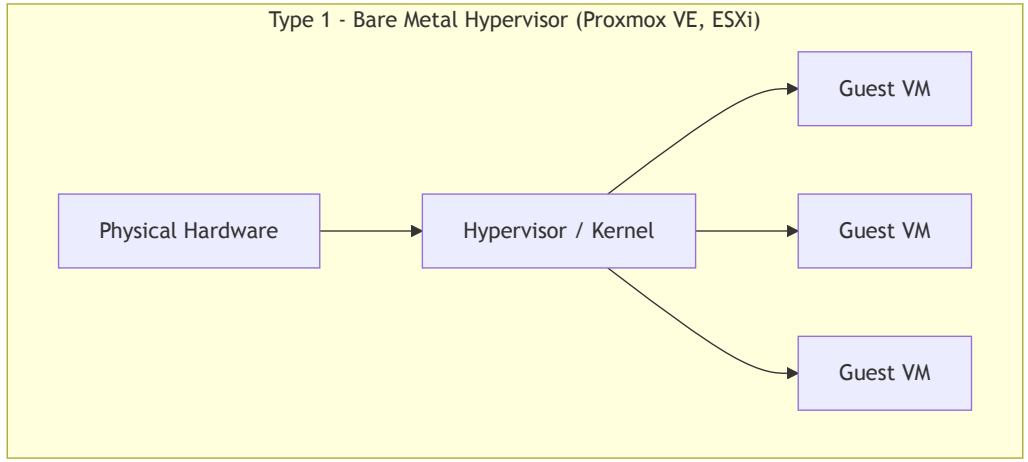
Area	VMware vSphere	Proxmox VE
Containers	Not native (Tanzu separate)	Native LXC on the same platform
Backup ecosystem	VADP / third-party	Proxmox Backup Server (PBS) + VADP-compatible tools
Licensing	Per-core subscription (Broadcom)	Open source + optional per-node subscription

The biggest mental shift: **there is no separate "vCenter appliance" to size, patch, and protect.** Every Proxmox node can manage the whole cluster because all nodes share `/etc/pve`.

3. Hypervisors: ESXi, KVM & LXC

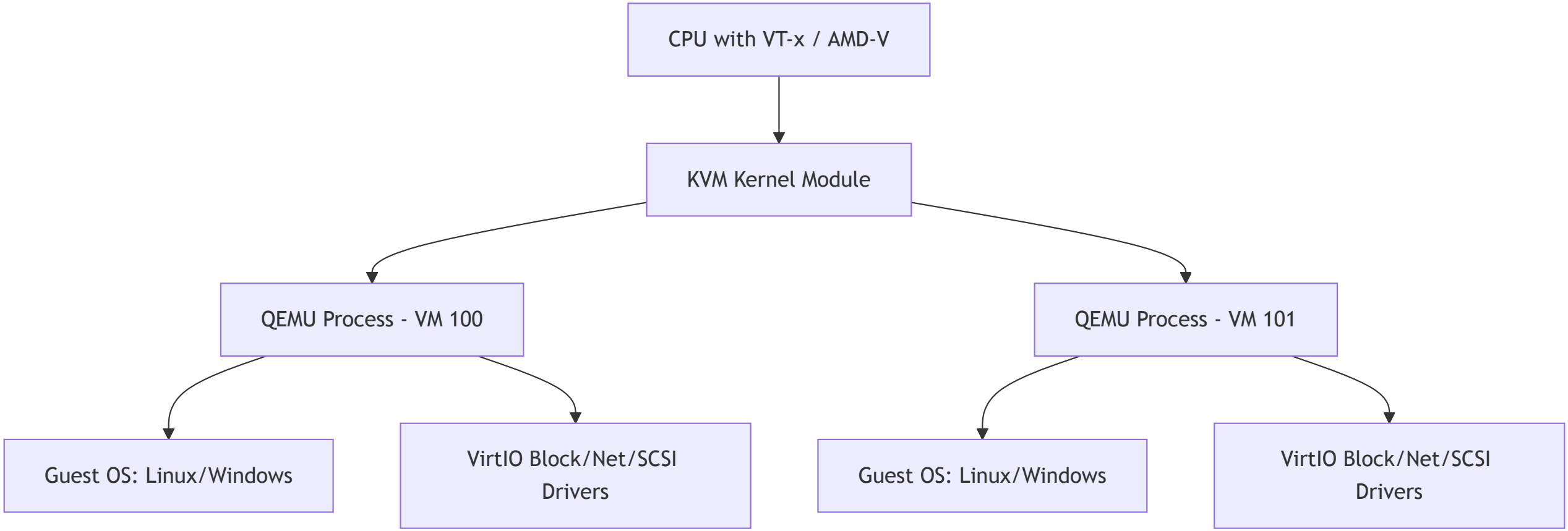
- **ESXi** is a Type 1 hypervisor: a purpose-built VMkernel with a small management footprint. VMs run as `vmx` processes scheduled by the VMkernel.
- **KVM** is a Linux kernel module that turns the Linux kernel itself into a hypervisor. Each VM is a normal QEMU process scheduled by the Linux scheduler.
- **LXC** is *not* a hypervisor at all — it is OS-level virtualization using kernel namespaces and cgroups. Containers share the host kernel.


Type 2 - Hosted Hypervisor (VirtualBox, VMware Workstation)
Physical Hardware
Host Operating System
Hypervisor Application
Guest VM
Guest VM


Type 1 - Bare Metal Hypervisor (Proxmox VE, ESXi)
Physical Hardware
Hypervisor / Kernel
Guest VM
Guest VM
Guest VM

Property	ESXi VM	KVM VM (Proxmox)	LXC Container (Proxmox)
Kernel	Guest kernel	Guest kernel	Host kernel shared
Hardware emulation	Full (VMware virtual devices)	Full (QEMU devices, usually VirtIO)	None
OS support	Almost any OS	Almost any OS	Linux only
Boot time	Seconds to minute	Seconds to minute	Sub-second to seconds
Density	Moderate	Moderate	High
Isolation strength	Strong (hardware boundary)	Strong (hardware boundary)	Weaker (shared kernel)

Rule of thumb for migration: migrated VMware VMs stay VMs. Do not "convert" them to containers as part of the migration — that is a separate application modernization project.


CPU with VT-x / AMD-V
KVM Kernel Module
QEMU Process - VM 100
QEMU Process - VM 101
Guest OS: Linux/Windows
VirtIO Block/Net/SCSI Drivers
Guest OS: Linux/Windows
VirtIO Block/Net/SCSI Drivers

QEMU/KVM Stack on a Proxmox Node

Each running VM is literally one `qemu-system-x86_64` process. That is why `qm showcmd <vmid>` is so useful during migration: it prints the exact command line Proxmox uses to start the guest, which you can compare with the VMware configuration.

4. Management & Operating Models

Concept	vSphere	Proxmox VE
Central management	vCenter (single appliance, SPOF if not HA)	Any node; configuration replicated
CLI	`esxcli`, PowerCLI	`pvesh`, `qm`, `pct`, `pvesm`, `pvecm`
API	vSphere REST/SOAP	REST (JSON) on every node, port 8006

Concept	vSphere	Proxmox VE
Configuration store	vCenter DB + host local	`/etc/pve` (cluster filesystem)
Automation	Terraform vSphere provider, Ansible	Terraform Proxmox provider, Ansible modules
Change propagation	vCenter pushes to hosts	Nodes read shared config; restart services as needed

```
# Inspect the whole cluster from any node
pvecm status
pvesh get /cluster/resources

# Manage a VM by VMID from any node
qm config 100
qm start 100
```

SSH access is first-class: hypervisor hosts are normal Debian systems. Plan SSH key distribution, sudo/role separation, and bastion access the same way you would for any Linux server fleet.

5. Licensing, Support & TCO

- Proxmox VE is **open source** (AGPLv3). You can run it without paying, including in production.
- An optional **subscription** (Basic/Standard/Premium) unlocks the enterprise repository, support tickets, and update guarantees.
- Without a subscription, use the `pve-no-subscription` repository (community-tested updates) and community forums.
- VMware's post-Broadcom licensing is **per-core subscription** with minimum core counts; many organizations are re-evaluating TCO and support models.

TCO discussion points for your business case:

Factor	Questions to answer
License cost	Cores, sockets, vCenter, vSAN, NSX, Aria, backup
Support	Response time, named contacts, escalation
Staff skills	vSphere admins → Linux/KVM/Proxmox training
Migration effort	Assessment, import, testing, cutover, rollback
Risk	Downtime windows, data loss, performance regression
Exit strategy	Open formats (qcow2/raw), documented configs

6. Realistic Performance Expectations

Be honest with stakeholders *before* migration. Proxmox VE is not automatically faster or slower than vSphere; outcomes depend on configuration.

Area	Expectation	Notes
CPU	Near-native with `host` CPU type	Named CPU models (`x86-64-v2-AES`) trade some performance for live-migration portability
Memory	Near-native	Watch ballooning and NUMA; size hosts with headroom
Storage IOPS/latency	Depends entirely on backend	ZFS with ARC can be excellent; misconfigured ZFS (no SLOG, wrong recordsize) can disappoint

Area	Expectation	Notes
Network throughput	Near line rate with VirtIO	Multiqueue, offloads, and bridge configuration matter
Live migration	Seconds of stun for memory copy	Depends on RAM size, dirty rate, network
HA restart	VM reboot time	Proxmox HA restarts VMs on another node; it is not continuous availability
Load balancing	Manual / scripted	No built-in DRS-style automatic placement; plan capacity deliberately

Common performance pitfalls after migration:

- Windows guests booting from IDE/SATA instead of VirtIO SCSI
- Missing multiqueue on high-throughput VMs
- ZFS ARC consuming RAM needed by guests
- Storage on a single spindle or USB disk in a "test" that becomes production

7. Proxmox VE Web UI, CLI & SSH Access

Everything in the web UI is a thin client over the **same REST API** used by the CLI.

Tool	Purpose
`pvesh`	Generic REST API client (`get` / `set` / `create` / `delete`)
`qm`	Manage QEMU/KVM virtual machines
`pct`	Manage LXC containers
`pvesm`	Manage storage
`pvecm`	Manage cluster membership
`pveceph`	Manage hyper-converged Ceph
`pveum`	Manage users, realms, ACLs

```
# Same operation, three ways
qm list
pvesh get /nodes/pve1/qemu
curl -k -H "Authorization: PVEAPIToken=..." https://pve1:8006/api2/json/nodes/pve1/qemu
```

SSH hygiene for a migration project:

```
# Distribute an admin key to all nodes
ssh-copy-id root@pve1.example.com
# Prefer a dedicated admin user + sudo on production nodes
```

Login to `https://<node-ip>:8006` with `root@pam` (or your realm user). Keep credentials in a password manager; for automation use **API tokens** with privilege separation.

8. Host, VM & Container Management — Limits of Containers

Proxmox VE manages three object types: **nodes** (hosts), **VMs** (QEMU/KVM), and **containers** (LXC). Each has its own config file under `/etc/pve`.

```
ls /etc/pve/qemu-server/   # VM configs: 100.conf, 101.conf
ls /etc/pve/lxc/           # Container configs: 200.conf
cat /etc/pve/storage.cfg    # Cluster-wide storage definitions
```

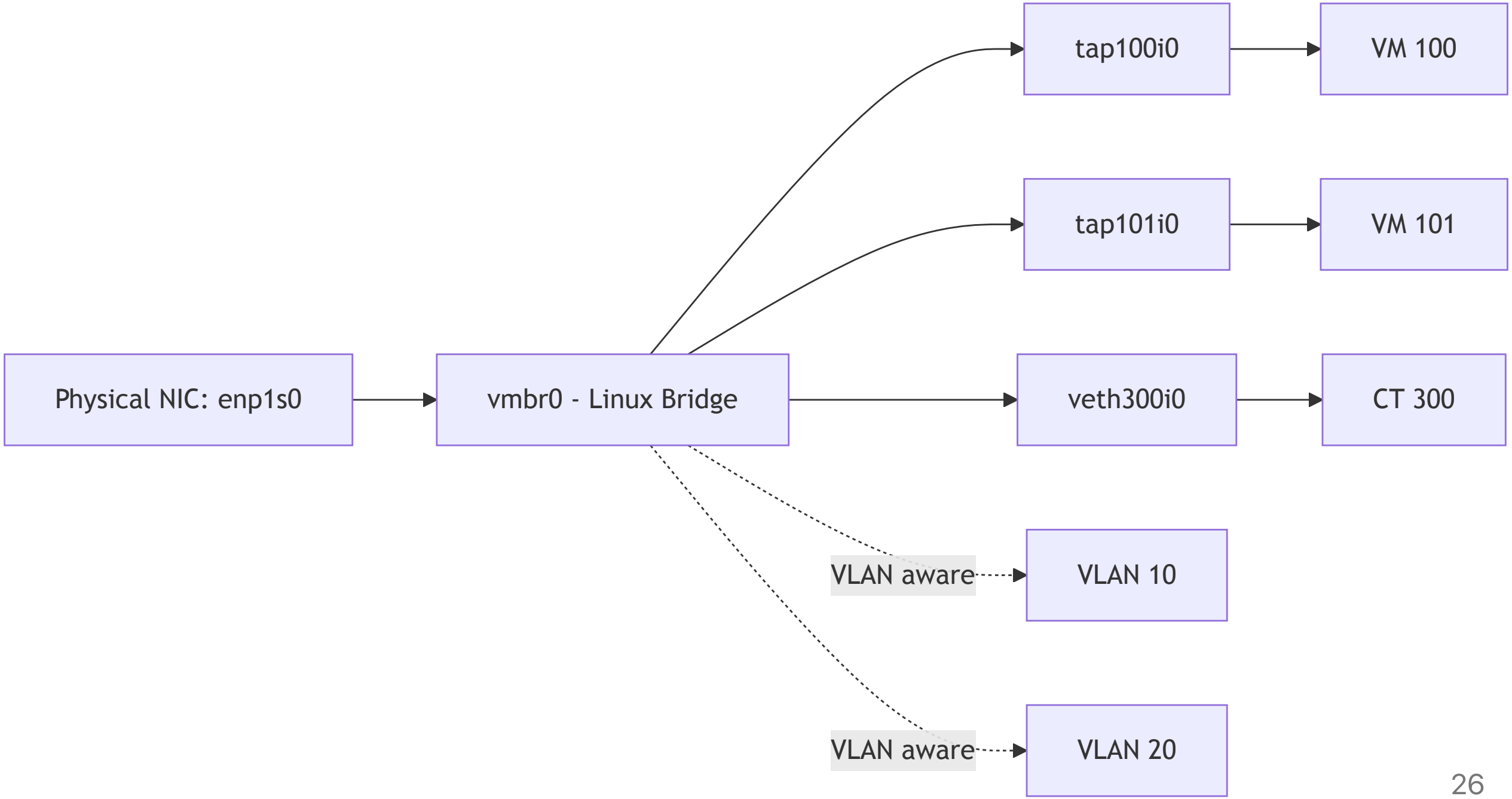
Where containers shine (and where they do not):

Use case	VM	LXC container
Migrated Windows Server	✅	❌ (Windows cannot run in LXC)
Migrated Linux appliance with custom kernel	✅	❌
New Linux microservice	✅	✅ (lighter, faster)
Security boundary between tenants	✅	⚠️ (shared kernel)
GPU/PCIe passthrough	✅	⚠️ (possible but more constrained)
Legacy kernel modules	✅	❌

Do not mix "migration" with "modernization." First get the VMware workload running unchanged on Proxmox. Optimize or containerize later, in a separate project with its own testing.

9. Networking Basics — Linux Bridges & VLANs

Proxmox VE networking builds on standard **Linux bridges** (`vmbr0` , `vmbr1` , …) — software switches connecting physical NICs, VM `tap` interfaces, and container `veth` pairs.


tap100i0
VM 100
tap101i0
VM 101
Physical NIC: enp1s0
vmbr0 - Linux Bridge
veth300i0
CT 300
VLAN aware
VLAN 10
VLAN aware
VLAN 20

```
# /etc/network/interfaces snippet
auto vmbr0
iface vmbr0 inet static
    address 10.10.10.5/24
    gateway 10.10.10.1
    bridge-ports enp1s0
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes
```

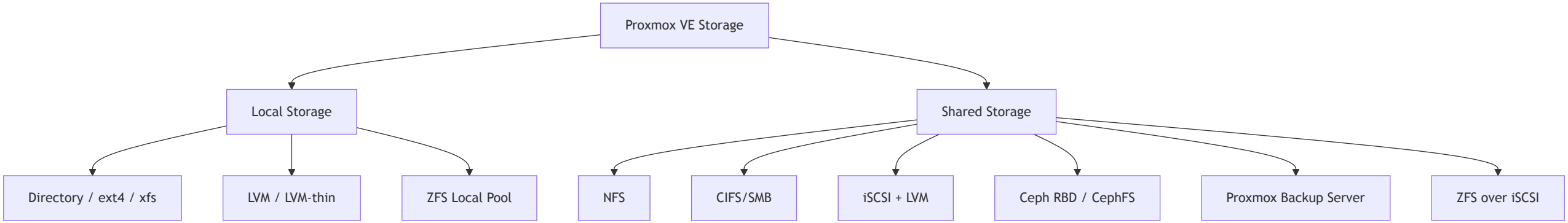
Migration mapping exercise (do this in the lab):

VMware concept	Proxmox equivalent	Notes
vSwitch / Distributed vSwitch	Linux bridge (`vmbr0`)	One bridge can be VLAN-aware
Port group (access VLAN)	Bridge + VLAN tag on NIC/vNIC	Tag on VM NIC or bridge
Trunk port group	VLAN-aware bridge	Multiple VLANs over one uplink
VMkernel port (management/vMotion)	Host IP on bridge or separate bridge	Keep management isolated
dvUplink / NIC teaming	Bond (`bond0`) as bridge port	LACP/active-backup modes

A **VLAN-aware bridge** lets a single `vmbr0` carry multiple tagged VLANs to different guests without one bridge per VLAN — this is the foundation for the SDN topics in the Architect course.

10. Storage Planning — Local vs. Shared

Proxmox VE separates **storage configuration** (`/etc/pve/storage.cfg` , cluster-wide) from the **actual backend technology**.


Proxmox VE Storage
Local Storage
Shared Storage
Directory / ext4 / xfs
LVM / LVM-thin
ZFS Local Pool
NFS
CIFS/SMB
iSCSI + LVM
Ceph RBD / CephFS
Proxmox Backup Server
ZFS over iSCSI

Storage type	Shared?	Snapshots	Typical migration use
Directory	No	qcow2 only	ISOs, templates, small labs
LVM-thin	No	Yes	Fast local VM disks on a single node
ZFS	No (local)	Yes	Resilient single-node storage, replication
NFS/CIFS	Yes	Depends on format	Shared ISO/template/backup storage, simple migration target
Ceph RBD/CephFS	Yes	Yes	HA clusters, scale-out, live migration with shared storage
Proxmox Backup Server	Yes (backup target)	N/A	Deduplicated backups, restore tests

Before importing anything, answer:

1. Where will migrated VM disks live (local ZFS, NFS, Ceph)?
2. Is the storage shared across all target nodes (required for live migration/HA)?
3. What are the IOPS/throughput and latency requirements per workload?
4. How much free space is needed for import *plus* snapshots/backups?
5. What is the backup target and retention policy?

11. VMware Inventory & Assessment

A migration succeeds or fails in the **assessment** phase. Collect facts before touching production.

Data to collect (per VM):

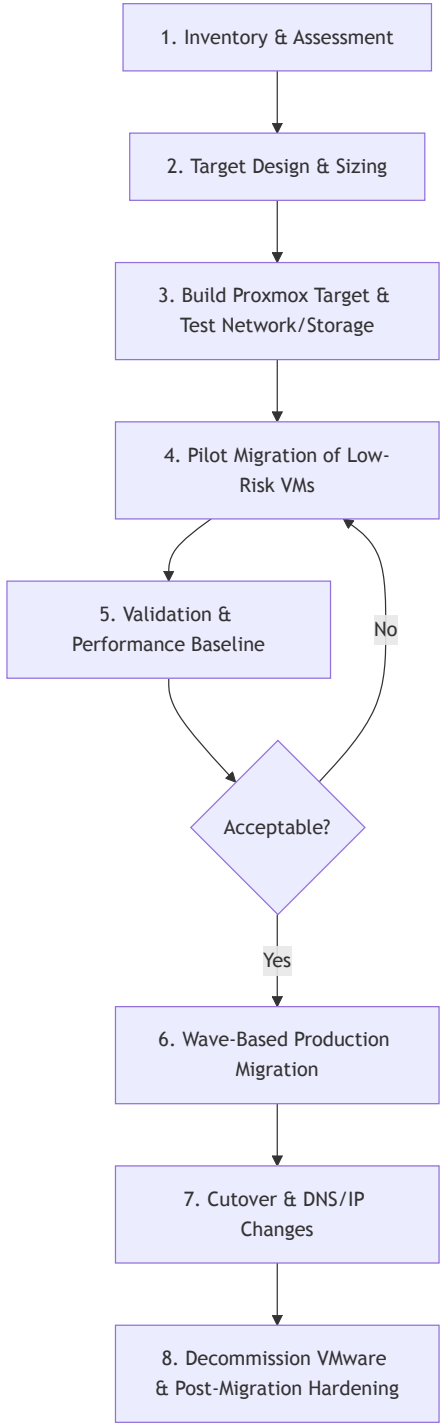
Field	Why it matters
VM name, UUID, guest OS	Identification and import mapping
vCPU, RAM, reservations/limits	Sizing target node and VM config
Disk count, size, thin/thick, RDM	Storage planning and import time
NIC count, port groups, VLANs, MAC	Network mapping and DHCP/DNS
VMware Tools version	Driver and guest agent strategy

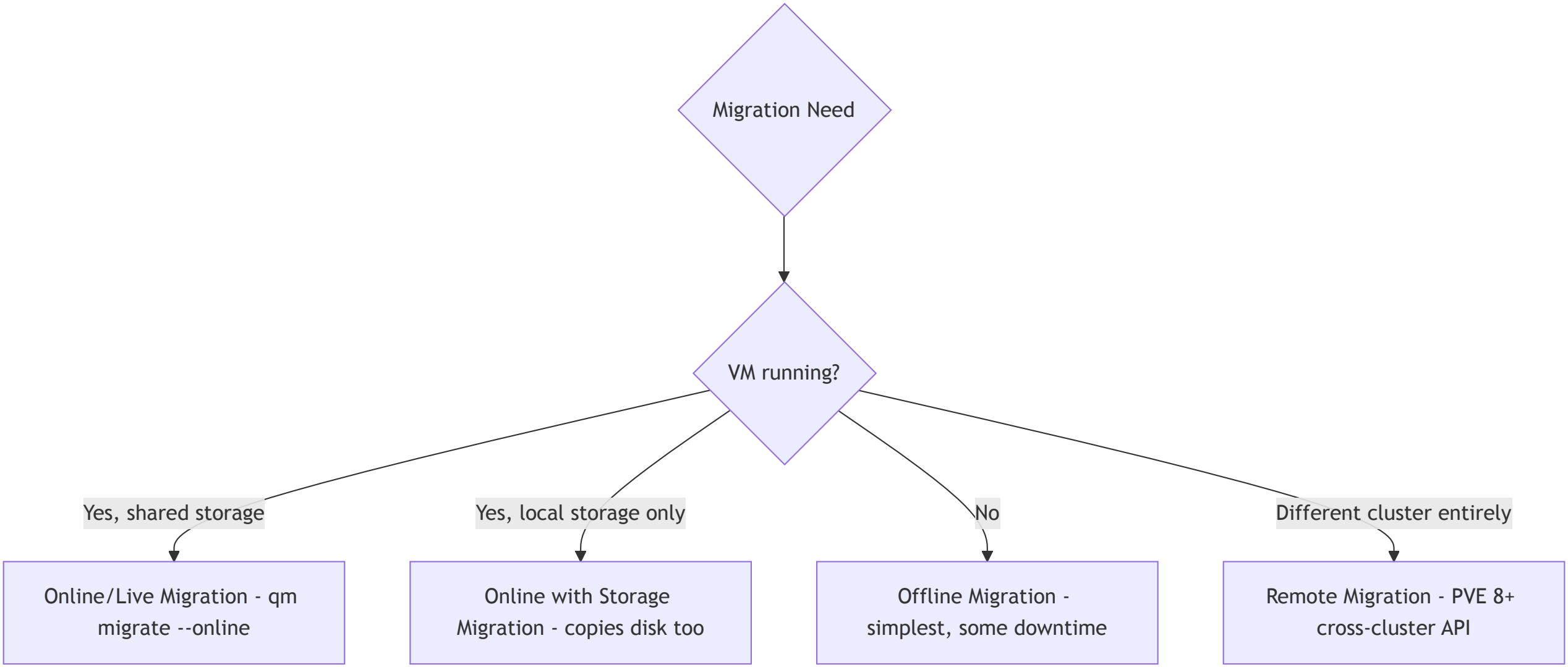
Field	Why it matters
Snapshots	Flatten or consolidate before import
Power state	Only powered-off VMs can be imported consistently
Dependencies	Boot order, app tiers, DB replication
Special features	vGPU, USB, serial, affinity/anti-affinity, DRS rules
Backup status	Last successful backup and restore test

Tooling: vCenter inventory, PowerCLI (`Get-VM`, `Get-HardDisk`, `Get-NetworkAdapter`), RVTools, or vendor assessment tools.

Classify each VM:

Class	Description	Migration approach
A — Straightforward	Linux/Windows on standard disks, no snapshots	Import early in waves
B — Needs adaptation	Special drivers, static IPs, clustered apps	Import with runbook
C — Complex	RDM, vGPU, MSCS, legacy OS	Redesign or defer
D — Do not migrate	Obsolete, replace with SaaS/container	Retire


1. Inventory & Assessment
2. Target Design & Sizing
3. Build Proxmox Target & Test Network/Storage
4. Pilot Migration of Low-Risk VMs
5. Validation & Performance Baseline
No
Acceptable?
Yes
6. Wave-Based Production Migration
7. Cutover & DNS/IP Changes
8. Decommission VMware & Post-Migration Hardening


Migration Need
VM running?
Yes, shared storage
Yes, local storage only
No
Different cluster entirely
Online/Live Migration - qm migrate --online
Online with Storage Migration - copies disk too
Offline Migration - simplest, some downtime
Remote Migration - PVE 8+ cross-cluster API

12. Migration Strategy, Waves & Test Systems

Wave planning principles:

- Start with **low-risk, non-production** VMs to learn the tooling
- Group by application tier and dependency, not just by size
- Keep wave size small enough to validate and roll back in one maintenance window
- Migrate monitoring/backup/identity infrastructure late, but test them early
- Never migrate the last copy of a critical service without a tested rollback

Test systems:

- Build an isolated Proxmox target (one node is enough for first tests) on a separate management network
- Use a test ESXi host or a small set of powered-off VMs
- Document every step in a runbook; turn the runbook into the production procedure
- Perform at least one **full restore test** from backup before cutover

Rollback plan (write it before you start):

1. Keep source VM untouched until acceptance
2. Keep source powered off but not deleted for a defined window (e.g. 2 weeks)
3. Document IP/DNS revert steps
4. Define "no-go" criteria and who can call them

13. Lab Environment Walkthrough

For the exercises in this course you will use:

- A source VMware environment (ESXi host or small vCenter) with sample Linux, Windows, and FreeBSD VMs
- A Proxmox VE target (single node or small cluster) with preconfigured storage and networking
- A management workstation with SSH, a browser, and (optionally) PowerCLI/RVTools
- Network access to both environments plus DNS/NTP

Before Day 2: confirm you can log into both vCenter/ESXi and the Proxmox web UI, run `qm list` and `pvesm status` on the target, and identify at least three candidate VMs (one Linux, one Windows, one FreeBSD or alternate Linux) for the import lab.

14. Migration Governance & Roles

A migration is a project, not just a technical task. Define ownership before the first import.

Role	Responsibility	Typical owner
Migration lead	Plan, schedule, go/no-go	Senior virtualization admin
VMware SME	Source inventory, snapshots, dependencies	vSphere admin
Proxmox SME	Target design, import, tuning	Linux/virtualization admin
Network engineer	VLANs, firewall, load balancers	Network team

Role	Responsibility	Typical owner
Storage engineer	Target storage, performance, capacity	Storage team
Application owner	Smoke tests, acceptance	App team
Security officer	Hardening, compliance, audit	Security team
Change manager	CAB, comms, blackout windows	ITSM

Governance Checklist

- [ ] Project charter with scope, timeline, budget, and success criteria
- [ ] RACI matrix published and agreed
- [ ] Weekly steering meeting with stakeholders
- [ ] Change advisory board (CAB) dates booked for each wave
- [ ] Communication plan for users and support desk
- [ ] Risk register reviewed at every steering meeting
- [ ] Single source of truth for migration tracker (spreadsheet/ticket system)

Success Criteria & KPIs

KPI	Target	Measurement
VMs migrated per wave	100% of wave scope	Migration tracker
Import success rate	≥ 95% first attempt	Task logs
Post-migration incidents	0 Sev-1 in 2 weeks	Ticketing
Performance regression	≤ 10% vs baseline	Benchmark report
Backup success rate	100%	PBS reports
Rollback usage	0 (ideal) but tested	Drill records
User-reported issues	≤ agreed threshold	Helpdesk

Communication Plan

Audience	Message	Channel	Timing
Executives	Status, risks, budget	Steering deck	Weekly
IT teams	Technical changes, firewall, IP	Email + wiki	Per wave
Application owners	Downtime windows, test steps	Direct meeting	2 weeks ahead
End users	Service availability	Intranet/email	1 week ahead
Support desk	Known issues, escalation	Runbook	3 days ahead

15. VMware Inventory Deep Dive

Inventory quality determines migration success. "I think that VM has 2 vCPUs" is not an inventory.

Minimum viable inventory fields:

```
vm_name, uuid, guest_os, power_state, vcpu, ram_mb, disk_count,
disk_size_gb, disk_type(thin/thick), datastore, snapshot_count,
nic_count, portgroup, vlan, mac, vm_tools_version, backup_job,
owner, tier(critical/standard/dev), notes
```

PowerCLI Inventory Script (Example)

```
Connect-VIServer vcenter.example.com -Credential (Get-Credential)
Get-VM | Select-Object Name, PowerState,
  @{N='vCPU';E={$_.NumCpu}},
  @{N='RAM_GB';E={[math]::Round($_.MemoryGB,2)}},
  @{N='Disk_GB';E={[math]::Round(($_.HardDisks | Measure-Object -Property CapacityGB -Sum).Sum,2)}},
  @{N='SnapshotCount';E={(Get-Snapshot -VM $_).Count}},
  @{N='PortGroups';E={($_.NetworkAdapters | ForEach-Object {$_.NetworkName}) -join ';'}},
  @{N='Tools';E={$_.ExtensionData.Guest.ToolsVersion}} |
  Export-Csv -NoTypeInformation vm-inventory.csv
```

Dependency Mapping

Migrate by **dependency graph**, not by VM size.

- Identify boot order within application tiers
- Map database → app server → web server → load balancer chains
- Note shared services (DNS, AD, NTP, monitoring, backup)
- Identify VMs that must move together in one window
- Mark "keystone" VMs (e.g., domain controllers) for late, careful migration

```
Tier 0: DNS, NTP, AD, monitoring
Tier 1: Databases
Tier 2: Application servers
Tier 3: Web/API frontends
Tier 4: Dev/test and batch
```

Snapshot Consolidation

Snapshots are the #1 cause of failed or inconsistent imports.

- List snapshots per VM (`Get-Snapshot`)
- Delete unneeded snapshots and consolidate chains
- Verify no snapshot older than your policy (e.g., 72 hours)
- Export a "clean" VMDK after consolidation
- Re-run inventory after consolidation to update sizes

```
Get-VM | Get-Snapshot | Where-Object {$_.Created -lt (Get-Date).AddDays(-3)} |
  Select-Object VM, Name, Created, SizeGB
```

16. Capacity & Sizing Calculations

CPU sizing example:

- Source: 40 VMs × 4 vCPU = 160 vCPU
- Target ratio 3:1 → 54 pCPU required
- Add 20% HA headroom → 65 pCPU
- Choose 4 nodes × 16 cores = 64 pCPU → add 1 node or reduce ratio

Memory sizing example:

- Sum guest RAM: 40 × 8 GB = 320 GB
- ZFS ARC: 16 GB per node × 4 = 64 GB
- Host overhead: 10% = 32 GB
- Total: 416 GB across cluster (104 GB/node)

Storage Capacity Planning

```
Required = (sum VM disk sizes × growth factor) + snapshot reserve + backup staging
Example:  (2 TB × 1.3) + 0.5 TB + 0.5 TB = 3.6 TB usable
With RAIDZ2 ~ 60% efficiency → 6.0 TB raw
```

- Plan for 12–18 months of growth, not just today
- Separate performance tiers (SSD for DB, HDD for archive)
- Include backup target capacity (PBS dedup often 2–5× logical)
- Leave ≥ 20% free space for ZFS/Ceph health

Network Bandwidth Planning

Traffic	Estimate	Notes
VM production	1–10 Gbps	Depends on app
Live migration	1–25 Gbps	RAM size × dirty rate
Storage replication	1–10 Gbps	ZFS/Ceph sync
Backup to PBS	1–10 Gbps	Dedup reduces over time
Management	100 Mbps–1 Gbps	Keep isolated

Rule: migration traffic must not starve production. Use a dedicated migration VLAN or schedule imports off-peak.

17. Network Design Examples

Example VLAN plan:

VLAN	Name	Subnet	Purpose
10	MGMT	10.10.10.0/24	Proxmox UI/API, SSH
20	VM-PROD	10.20.0.0/16	Migrated production VMs
30	VM-DEV	10.30.0.0/16	Dev/test VMs
40	STORAGE	10.40.0.0/24	NFS/iSCSI/Ceph
50	BACKUP	10.50.0.0/24	PBS traffic
99	MIGRATION	10.99.0.0/24	Temporary import traffic

Bridge Configuration Example

```
# /etc/network/interfaces
auto enp1s0
iface enp1s0 inet manual

auto bond0
iface bond0 inet manual
    bond-slaves enp1s0 enp2s0
    bond-miimon 100
    bond-mode 802.3ad

auto vmbr0
iface vmbr0 inet static
    address 10.10.10.5/24
    gateway 10.10.10.1
    bridge-ports bond0
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes
    bridge-vids 2-4094
```

DNS & DHCP Planning

- Keep DNS records updated during cutover (short TTLs in advance)
- Decide: keep IPs (simpler) vs. renumber (cleaner, more work)
- Document MAC→IP mappings for static assignments
- Test reverse DNS for migrated VMs
- Consider DHCP reservations for VMs that previously had static IPs

```
# Shorten TTL before migration
$TTL 300
web01  IN  A  10.20.0.10
```

18. Storage Design Examples

ZFS pool example:

```
zpool create -f -o ashift=12 tank raidz2 \
  /dev/disk/by-id/ata-... /dev/disk/by-id/ata-... ...
zfs create tank/vmdata
zfs set compression=lz4 tank/vmdata
zfs set recordsize=64K tank/vmdata
zfs set atime=off tank/vmdata
```

- `ashift=12` for 4K sector disks
- `recordsize` 64K for VMs (match workload)
- `compression=lz4` (cheap, beneficial)
- Reserve 20% free space

IOPS Planning

Workload	IOPS	Notes
Idle VM	10–50	Baseline
Web server	100–500	Random read
Database (OLTP)	1,000–10,000+	Latency-sensitive
Backup target	Sequential high	Throughput over IOPS

Rule: measure source IOPS with `iostat -x 1` during peak before sizing target storage.

19. Identity & Access Planning

- Map vSphere roles (Administrator, Read-only, VM Power User) to Proxmox roles
- Decide realm strategy: local `pve` vs. AD/LDAP/OIDC
- Plan 2FA rollout with a pilot group
- Create break-glass local admin account (documented, secured)
- Inventory all automation accounts and their required permissions
- Plan API token rotation schedule

Break-Glass Account

```
# Create a local emergency admin (store credentials in vault)
pveum user add breakglass@pve --password '***'
pveum acl modify / --users breakglass@pve --roles Administrator
# Disable interactive login normally; enable only during incidents
```

- Store in a sealed envelope / vault
- Test quarterly
- Alert on use
- Rotate after any use

20. Backup & DR Planning Before Migration

- Choose backup target (PBS recommended)
- Size PBS: dedup ratio often 2–5×, but plan conservative
- Define RPO/RTO per tier
- Document restore procedure and test it
- Decide encryption and key custody
- Plan off-site replication of PBS datastore

RPO/RTO Matrix (Example)

Tier	RPO	RTO	Method
Tier 0 (identity)	15 min	1 h	Replication + HA
Tier 1 (DB)	1 h	4 h	PBS hourly + restore test
Tier 2 (app)	4 h	8 h	PBS daily
Tier 3 (web)	24 h	24 h	PBS daily
Dev/test	7 days	7 days	Weekly

21. Risk Register & Rollback Strategy

Risk	Impact	Likelihood	Mitigation
Import failure mid-copy	High	Medium	Free space check, retry, resume
Guest won't boot	High	Medium	Firmware match, rescue ISO, rollback
Performance regression	Medium	Medium	Baseline, tuning, rollback
Network misconfig	High	Medium	Test in isolated VLAN first
Data loss	Critical	Low	Keep source VM, tested backups
Scope creep	Medium	High	Change control, wave discipline

Rollback Timeboxes

- Define rollback decision point in every runbook (e.g., "if not healthy by T+2h, roll back")
- Assign rollback authority (who can call it)
- Keep source VM powered off but intact for 2–4 weeks
- Document DNS/IP revert steps
- Test rollback at least once in a lab

22. Migration Wave Planning Workbook

Wave	VMs	Window	Owner	Go/No-Go	Rollback
0	3 test VMs	Week 1	Migration lead	Lead	Keep source
1	10 low-risk Linux	Week 2	VMware SME	CAB	24h
2	10 Windows apps	Week 3	App owners	CAB	48h
3	Databases	Week 4	DBA	CAB	72h
4	Identity/monitoring	Week 5	Infra	CAB	72h
5	Remaining	Week 6+	Migration lead	CAB	per VM

Go/No-Go Gate

Before each wave, verify:

- [ ] Target capacity confirmed
- [ ] Backups tested
- [ ] Network/firewall changes applied
- [ ] Application owners available
- [ ] Rollback tested
- [ ] Monitoring in place
- [ ] Communication sent
- [ ] CAB approved

23. Assessment Lab Deliverable

Deliverable: a signed-off migration assessment package containing:

1. Inventory spreadsheet (all VMs)
2. Dependency map
3. Target architecture (network + storage + identity)
4. Capacity plan with math
5. Wave plan with dates
6. Risk register + rollback plan
7. Backup/DR plan
8. Success criteria & KPIs

Review with stakeholders before Day 2.

24. Appendix A — Command Reference (Proxmox)

```
# Cluster
pvecm status
pvecm create <name>
pvecm add <host>

# VMs
qm list
qm config <vmid>
qm start|stop|shutdown|reset <vmid>
qm migrate <vmid> <node> [--with-local-disks] [--online]

# Storage
pvesm status
pvesm add <type> <id> [options]

# Users
pveum user add <user@realm>
pveum acl modify <path> --users <u> --roles <r>
```

Appendix B — Network Commands

```
ip a
ip r
bridge vlan show
bridge link
cat /etc/network/interfaces
systemctl restart networking
ping -c 4 <gateway>
traceroute <host>
```

Appendix C — Storage Commands

```
pvesm status
lsblk
zpool status
zfs list
zfs get compressratio,recordsize,used,avail tank/vmdata
iostat -x 1 5
fio --name=test --ioengine=libaio --iodepth=32 --rw=randread --bs=4k --direct=1 --size=1G --runtime=60
```

Appendix D — Assessment Templates

- Inventory CSV template (see section 15)
- Dependency map (tiered list)
- Capacity spreadsheet (CPU/RAM/storage/network)
- Wave plan (table)
- Risk register
- Runbook template (steps, owner, rollback)
- Test matrix (function + performance)

Store templates in version control and reuse across projects.

Appendix E — Frequently Asked Questions

Q: Can we migrate powered-on VMs?

A: The ESXi import path is cold. For minimal downtime, use snapshot-based or replication-based strategies.

Q: Do we need shared storage before migration?

A: Not for single-node targets. For HA/live migration, yes.

Q: How long does an import take?

A: Roughly disk size ÷ available bandwidth + overhead. A 100 GB VM over 1 Gbps ≈ 15–20 minutes plus conversion.

Q: Can we automate imports?

A: Yes, via `qm importdisk` and scripted config; the UI wizard is best for learning.

Q: What about vGPU VMs?

A: Treat as complex (Class C); plan passthrough and licensing separately.

25. Migration Cost Model

Cost category	VMware (annual)	Proxmox (annual)
Hypervisor license	Per-core subscription	Optional subscription
vCenter	Included/extra	None (distributed)
Backup (VADP)	Third-party	PBS (open source)
Storage	vSAN license	Ceph/ZFS (open source)
Training	vSphere skills	Linux/KVM skills
Migration project	One-time	One-time

Build a 3-year TCO with conservative assumptions and a sensitivity analysis.

26. Hypervisor Market Comparison

Platform	Type	Strengths	Considerations
Proxmox VE	Type 1 (KVM)	Open source, integrated, simple	Linux skills required
VMware vSphere	Type 1	Mature ecosystem, DRS	Licensing cost, Broadcom
Hyper-V	Type 1	Windows integration	Less Linux-friendly
Nutanix AHV	Type 1	HCI simplicity	Vendor lock-in
KVM (manual)	Type 1	Flexible	More DIY

27. Storage Benchmark Examples

```
# Random read IOPS
fio --name=randread --ioengine=libaio --iodepth=64 \
  --rw=randread --bs=4k --direct=1 --size=4G --runtime=120 \
  --numjobs=4 --group_reporting

# Sequential throughput
fio --name=seqwrite --ioengine=libaio --iodepth=32 \
  --rw=write --bs=1M --direct=1 --size=8G --runtime=120
```

Record IOPS, bandwidth, and latency percentiles (p95/p99).

28. Network Troubleshooting Playbook

1. Verify link state: `ip link show`
2. Verify bridge membership: `bridge link`
3. Verify VLAN tags: `bridge vlan show`
4. Capture traffic: `tcpdump -ni vmbr0`
5. Test L3: `ping` , `traceroute`
6. Check firewall counters
7. Verify DNS/NTP
8. Compare with source VMware port group

29. Security Baseline for Migration Project

- Separate migration network from production
- Use dedicated service accounts (least privilege)
- Store credentials in a vault
- Enable 2FA for all operators
- Log all migration actions
- Scan transferred disks for malware
- Remove temporary access after cutover

30. Change Management Process

Step	Output
Request	RFC with scope, risk, rollback
Review	CAB approval
Implement	Runbook execution
Verify	Test results
Close	Sign-off, documentation

No production change without an approved RFC.

31. Documentation Standards

- Use Markdown in Git for all runbooks
- Include owner, date, version
- Add diagrams (Mermaid) for architecture
- Keep commands copy-pasteable
- Record deviations and lessons learned
- Review docs after each wave

32. Training Plan

Audience	Topics	Format
VMware admins	Proxmox UI/CLI, Linux basics	3-day workshop
Network team	Bridges, VLANs, SDN	1-day workshop
Storage team	ZFS/Ceph/PBS	2-day workshop
App owners	VM lifecycle, snapshots	Half-day
Ops team	Monitoring, backup, HA	2-day workshop

33. Lab: Build Target from Scratch

1. Install Proxmox VE on a test node
2. Configure repositories and updates
3. Create a Linux bridge with VLAN awareness
4. Create a ZFS pool and storage
5. Create a test VM and boot it
6. Configure a PBS backup target
7. Run a backup and restore

34. Lab: Assess a Sample VMware Environment

Given a small vSphere lab:

1. Run PowerCLI inventory
2. Classify VMs (A/B/C/D)
3. Build dependency map
4. Create wave plan
5. Estimate capacity
6. Present assessment to stakeholders

35. Lab: Design Review

Peer-review another team's assessment:

- Is inventory complete?
- Are dependencies captured?
- Is capacity math correct?
- Is rollback realistic?
- Are risks identified?

36. Appendix: Proxmox UI Map

Menu	Purpose
Datacenter	Cluster-wide settings, users, storage, backup
Node	Host status, shell, network, disks
VM/CT	Guest config, console, snapshots
Storage	Content, upload ISO, backups
Tasks	Operation logs
Replication	ZFS replication jobs

37. Appendix: Useful Links

- Proxmox VE documentation
- Proxmox Backup Server documentation
- Proxmox forum and wiki
- Debian administrator's handbook
- ZFS on Linux documentation
- Ceph documentation

38. Appendix: Sample Inventory CSV

```
vm_name,guest_os,vcpu,ram_gb,disk_gb,nic,portgroup,owner,tier
web01,Ubuntu 24.04,2,4,40,1,VM-PROD,web-team,2
db01,Rocky 9,4,16,200,1,VM-PROD,dba-team,1
```

39. Appendix: Capacity Spreadsheet Columns

- VM name, tier
- vCPU, RAM (GB)
- Disk (GB), IOPS
- Network (Mbps)
- Source host
- Target node
- Wave
- Status

40. Appendix: Wave Plan Template

Wave	VMs	Window	Owner	Go/No-Go	Rollback
0	3 test	Week 1	Lead	Lead	Keep source
1	10 Linux	Week 2	SME	CAB	24h

41. Appendix: Risk Register Template

ID	Risk	Impact	Likelihood	Mitigation	Owner
R1	Import failure	High	Medium	Retry, rollback	Lead

42. Appendix: Runbook Template

1. Purpose
2. Scope
3. Prerequisites
4. Steps (numbered, with commands)
5. Validation
6. Rollback
7. Owner and approver
8. Revision history

43. Appendix: Test Matrix Template

Test	Method	Pass criteria	Owner	Result
Boot	Start VM	Login prompt	Lead	
Network	Ping/DNS	All pass	Lead	

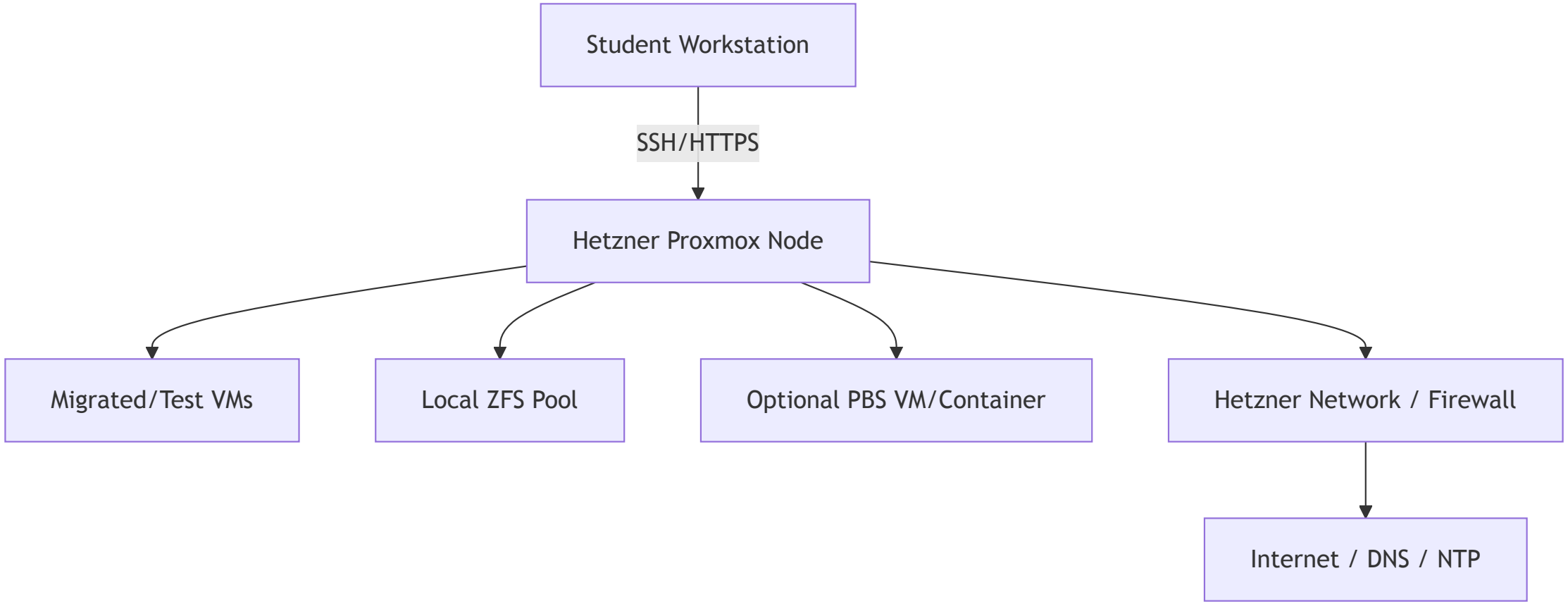
44. Appendix: Performance Baseline Template

VM	CPU	Mem	Disk IOPS	Disk lat	Net	Date

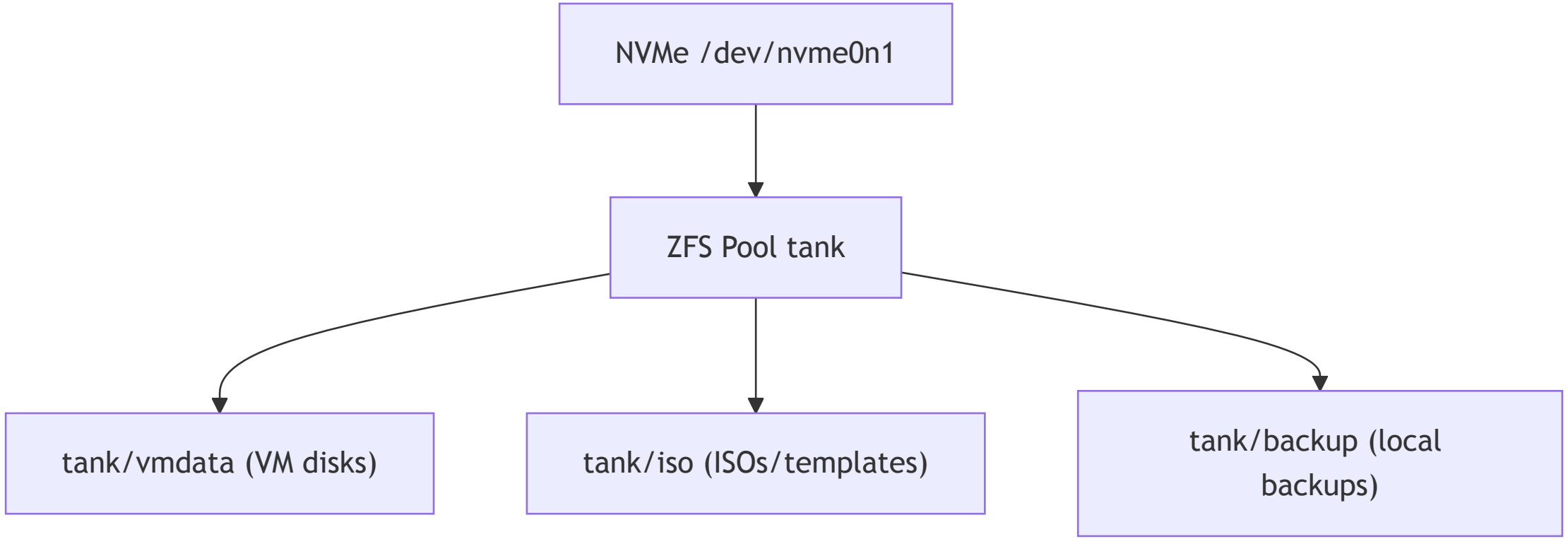
45. Appendix: Migration Tracker Fields

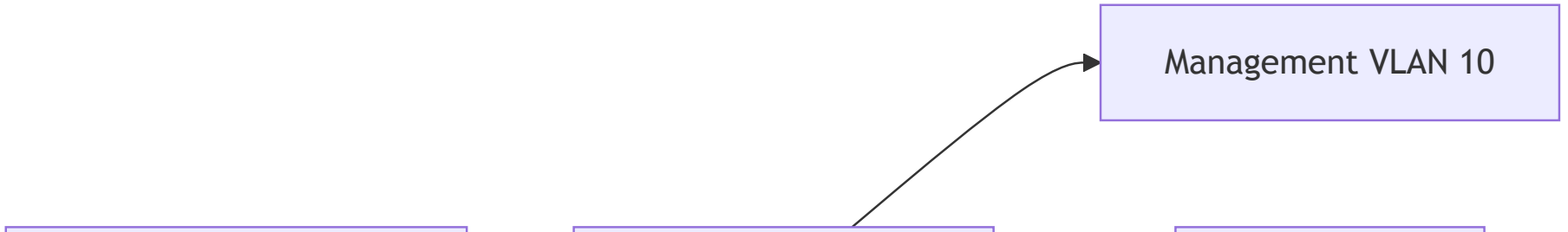
- VM name, UUID
- Source host, target node
- Wave, status
- Import start/end
- Issues found
- Fixes applied
- Test results
- Go/no-go
- Cutover date
- Notes

Hetzner Lab Topology (Reference)



Hetzner Storage & Network Layout (Reference)





Day 1 — Key Takeaways

- Proxmox VE is a **distributed, open-source** Type 1 platform; there is no separate vCenter to operate
- Migrated VMware VMs **remain VMs** — containerization is a later, separate decision
- **Assessment and wave planning** dominate migration success more than the import tool itself
- Networking and storage must be **designed and tested** before the first production VM is imported
- Always keep a **tested rollback path** and do not delete source VMs prematurely
- Realistic performance expectations prevent stakeholder surprises after cutover